



Report

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Summary

① Tasks Status

② Results

③ Selected Article

General Tasks Status

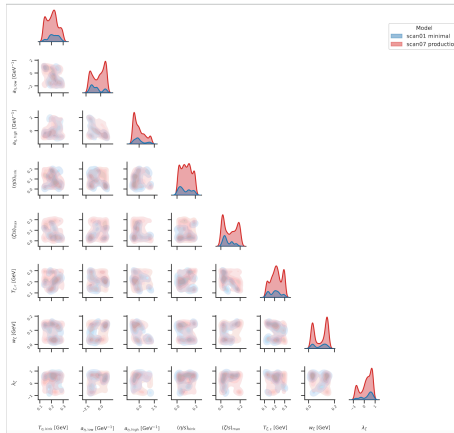
Task	Status	
2026-06 RETINHA presentation		Link
2026-06 Scientific report		Link

Task	Status	
Journal Club - Automations		

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Mcmc Params	✓	Plots
↪ RDF Uncertainty Correlation	✓	Plots
↪ CS Uncertainty Correlation	✓	Plots
↪ None Uncertainty Correlation	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation	🔄	
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation	🔄	

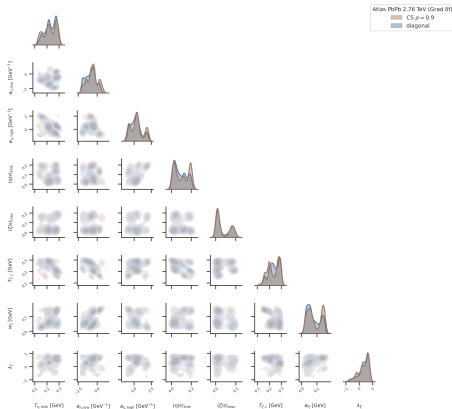
Mcmc params study



Figure

Uncertainty correlation study

Bayesian uncertainty-correlation study: CS vs diagonal



Figure

Selected Article

Hadronic tensor in lattice gauge theories by quantum computing

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The hadronic tensor encodes crucial information regarding the internal structure of hadrons, reflecting the non-perturbative features of quantum chromodynamics (QCD). In this work, we directly compute the hadronic tensor within (1+1)-dimensional U(1) and SU(2) gauge theories by evaluating real-time current-current correlation functions. Utilizing quantum algorithms executed on classical hardware, we demonstrate that the hadron form factors for both meson and baryon states can be reliably extracted from the hadronic tensor. Our methodology is validated by strong agreement with both direct calculation and exact diagonalization of the form factors.

I. INTRODUCTION

Figure: Lattice Quantum Chromo Dynamics (QCD) calculations on Quantum Computers

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